

A Ball-Convergent Subsequence Forces the Full Steiner Sequence to Converge

Candidate full solution of Problem 1 in arXiv:2102.07103

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Status. Candidate full proof, likely valid, pending human review. Problem 1 has an affirmative answer, with a quantitative Hausdorff estimate.

1 Source question and answer

Y. Lin, *The Petty projection inequality for sets of finite perimeter*, arXiv:2102.07103 [1], writes $\mathcal{C}^n$ for the compact sets of finite perimeter in $\mathbb{R}^n$ with nonempty interior. For $E \in \mathcal{C}^n$ and directions $u_i \in S^{n-1}$, let

$$E_i = S_{u_i} \cdots S_{u_1} E,$$

and let $E^* = \overline{B}_r$ be the origin-centered ball with $|E^*| = |E|$. Problem 1 asks whether convergence of one subsequence $E_{i_j} \rightarrow E^*$ in Hausdorff distance forces the entire sequence to converge.

6 Open problems

Problem 6.1 For $E \in \mathcal{C}^n$, if $E_i := S_{u_i} \cdots S_{u_1} E$ for a sequence of directions $\{u_i\}$ and there exists a subsequence E_{i_j} converges to E^* in the Hausdorff distance, then does E_i converge to E^* in the Hausdorff distance?

If the answer of Problem 6.1 is positive, then the sequence $\{E_i\}$ in Theorem 3.1 converges to E^* , which is stronger than the convergence of its subsequence.

Problem 6.2 For $E \in \mathcal{C}^n$, if $E_i := S_{u_i} \cdots S_{u_1} E$ for a sequence of directions $\{u_i\}$ and $E_i \rightarrow E^*$ in the Hausdorff distance, then does $P(E_i)$ converge to $P(E^*)$?

If the answer of Problem 6.2 is positive, then by Proposition 4.1, the sequence $\{\Pi E_i\}$ in Lemma 5.5 converges to ΠE^* , which is stronger than its convergence to a convex body K satisfying $\Pi E^* \subset K$.

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Figure 1: Problem 1 in the source PDF.

The answer is yes. The proof uses only volume preservation and a Lyapunov quantity that is especially transparent here: the radius of the smallest *origin-centered* ball containing the set.

2 Proof intuition

Steiner symmetrization preserves volume and cannot increase the centered circumradius. The ball-convergent subsequence forces that monotone radius to have the ball radius as its limit. Once every late set lies in an arbitrarily thin outer shell around the equal-volume ball, any

macroscopic hole inside the ball would have more volume than that shell can hold. This converts radius convergence into Hausdorff convergence for the whole sequence.

Theorem 1. *Let $E \in \mathcal{C}^n$, let (u_i) be any sequence of directions, and put $E_i = S_{u_i} \cdots S_{u_1} E$. If a subsequence converges in Hausdorff distance to $E^* = \overline{B}_r$, then*

$$E_i \longrightarrow E^* \quad \text{in Hausdorff distance.}$$

Thus Problem 1 in the source has an affirmative answer.

3 Equal volume upgrades outer-radius convergence

For a nonempty compact set A , write

$$R(A) = \max_{x \in A} |x|.$$

Lemma 1. *Let $A_k \subset \mathbb{R}^n$ be compact measurable sets with $|A_k| = |\overline{B}_r| > 0$. If $A_k \subset \overline{B}_{R_k}$ and $R_k \rightarrow r$, then $A_k \rightarrow \overline{B}_r$ in Hausdorff distance. More quantitatively, for all sufficiently large k ,*

$$d_H(A_k, \overline{B}_r) \leq \max \left\{ R_k - r, 2(R_k^n - r^n)^{1/n} \right\}. \quad (1)$$

Proof. Equal volume and $A_k \subset \overline{B}_{R_k}$ imply $R_k \geq r$. The outer half of the Hausdorff distance is at most $R_k - r$.

Set

$$\eta_k = \sup_{x \in \overline{B}_r} \text{dist}(x, A_k).$$

Fix $x \in \overline{B}_r$ and $0 < \varepsilon < \min\{\text{dist}(x, A_k), 2r\}$. Put

$$y = \left(1 - \frac{\varepsilon}{2r}\right)x.$$

Then

$$B(y, \varepsilon/2) \subset B(x, \varepsilon) \cap \overline{B}_r \subset \overline{B}_r \setminus A_k.$$

Consequently,

$$\omega_n(\varepsilon/2)^n \leq |\overline{B}_r \setminus A_k| = |A_k \setminus \overline{B}_r| \leq |\overline{B}_{R_k} \setminus \overline{B}_r| = \omega_n(R_k^n - r^n).$$

Since the last term tends to zero, $\eta_k < 2r$ for all sufficiently large k . Letting $\varepsilon \uparrow \text{dist}(x, A_k)$ and then taking the supremum over x gives $\eta_k \leq 2(R_k^n - r^n)^{1/n}$. This proves (1) and the lemma. $\square$

4 The centered circumradius is monotone

Lemma 2. *For every compact measurable $A \subset \mathbb{R}^n$ and every direction $u \in S^{n-1}$,*

$$R(S_u A) \leq R(A).$$

Proof. Let $R = R(A)$, so $A \subset \overline{B}_R$. Write a point as $x + tu$ with $x \in u^\perp$. The one-dimensional section of A above x is contained in

$$\left[-\sqrt{R^2 - |x|^2}, \sqrt{R^2 - |x|^2} \right].$$

Its length is therefore at most $2\sqrt{R^2 - |x|^2}$. Steiner symmetrization replaces that section by the centered interval of the same length, which is still contained in the displayed interval. Hence $S_u A \subset \overline{B}_R$. $\square$

Proof of Theorem 1. Steiner symmetrization preserves Lebesgue measure, so $|E_i| = |E| = |\overline{B}_r|$ for all i . Lemma 2 shows that

$$R(E_{i+1}) \leq R(E_i).$$

Moreover $R(E_i) \geq r$, since a set of volume $|\overline{B}_r|$ cannot fit inside a smaller centered ball.

The map $A \mapsto R(A)$ is 1-Lipschitz for the Hausdorff metric. Thus the assumed subsequential convergence gives $R(E_{i_j}) \rightarrow r$. The decreasing sequence $R(E_i)$ must consequently converge to r . Apply Lemma 1 with $A_k = E_k$ and $R_k = R(E_k)$ to obtain $E_i \rightarrow \overline{B}_r$ in Hausdorff distance. $\square$

5 Scope, verification, and Problem 2

1. The proof applies to arbitrary compact measurable starting sets of positive finite volume for which successive Steiner symmetrals are compact; finite perimeter is not used for Problem 1.
2. Null-set representatives cause no gap: compactness defines $R(A)$ and Hausdorff distance, while the volume comparison in Lemma 1 works for the chosen measurable representatives.
3. The case where x lies on the boundary of the ball is covered by the explicit inner ball $B(y, \varepsilon/2)$.
4. Formula (1) was checked independently on model sets consisting of a ball-volume compact set contained in a shrinking outer ball.

The source's Problem 2 asks in addition whether $P(E_i) \rightarrow P(E^\star)$ when the full sequence converges in Hausdorff distance. The argument above does not settle that question. Hausdorff convergence and equal volume alone do not control perimeter: arbitrarily fine boundary corrugations can retain excess perimeter. The monotonicity $P(E_{i+1}) \leq P(E_i)$ gives existence of a limit and lower semicontinuity gives $P(E^\star) \leq \lim_i P(E_i)$, but a new upper bound exploiting the iterative Steiner structure would still be required.

The local run indexes were searched for the arXiv identifier, both exact problem statements, and the core Steiner/Hausdorff keywords. A bounded external search on 11 August 2026 checked the exact wording and later work on convergence of symmetrization processes. It found broad convergence theorems and pathological direction sequences, but no explicit answer to this precise subsequence-to-full-sequence question for the source's compact finite-perimeter setting. This is not an exhaustive citation-graph review, so originality remains a candidate claim pending expert checking.

References

- [1] Y. Lin, The Petty projection inequality for sets of finite perimeter, arXiv:2102.07103 (2021), arXiv:2102.07103.